Central Limit Theorem

Central Limit Theorem

Let $X_1, X_2, \dots, X_n$ be identical and independent distributed random variables with $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$. We define

$$Y_n = \sqrt{n}\left(\frac{\bar{X} - \mu}{\sigma}\right) \text{ where } \bar{X} = x\frac{1}{n}\sum_{i=1}^{n} X_i.$$

Then, the distribution of the function Y_n converges to a standard normal distribution function as $n \to \infty$.

Central Limit Theorem

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Central Limit Theorem Proof

Example

Let $X_1, \dots, X_n \overset{iid}{\sim} \chi^2_p$, the MGF is $M(t) = (1-2t)^{-p/2}$. Find the distribution of $\bar{X}$ as $n \to \infty$.